

NATIONAL INSTITUTE OF TECHNOLOGY, KURUKSHETRA

Roll No.

Month and Year of Examination: Aug 2026

Programme: B. Tech. (PIE)

Subject: Value Engineering

Time: 3 HOURS

Semester: 4th

Max. Marks: 50

Paper Code: PIPC-214

Total no. of Questions: 06

Note: Attempt any five questions. All questions carry equal marks. It is **mandatory** to draw **CASH FLOW** diagram for all questions. All steps/computations must be clearly visible.

Q 1 a)	(i) Find the PW of an annuity of \$250 running from EOY 3 to EOY6, at annual return rate of 8% (ii) Find the FW of annuity of \$300 running from EOY 4 to EOY 8, annual return rate 9% at the EOY9. (iii) Find the net AW of investments of \$450 (EOY3) and \$600 (EOY6) and a revenue of \$1200 at EOY 8, taking $i=10\%$ and project life 8 years.	(2x3)
b)	The selling price of Airbus A380 jumbo jet was around \$430 million/unit and the production capacity of the plant has been around 27 units/year . The total variable cost per unit was \$250 million . There was a onetime investment of \$24 billion for design and development at the beginning of manufacturing. The total fixed costs were \$3600 million per year for the assembly line. Compute the breakeven point in number of years and no. of units to be sold .	(4)
Q2.	AMT Inc. is considering the purchase of an automated inspection system for initial investment of \$340,000 and the estimated market value after 09 years is \$65,000. Annual revenues attributable to the new system are \$1, 22,000, whereas additional annual expenses on maintenance and operation of the system will be \$34000. The firm's MARR is 15% per year. A major overhaul of the system will be required at end of year 4, at a cost of \$60000. Find the economical feasibility of project by Annual worth method. Also find the Capital recovery amount.	(7+3)
Q3.	a) Explain the concept of life cycle cost while showing the different phases and activities by drawing a neat figure and mention the important learnings from this concept point wise. b) Differentiate among direct costs, indirect costs and overhead costs in tabular form giving appropriate examples of each.	(6) (4)

Q4

The cost of a space craft is related to its weight, as per the data given below:

Space craft	Weight (lb)	Cost (\$M)
1	400	320
2	500	425
3	750	650
4	900	780
5	1200	1080
6	1500	1350

(10)

Establish a cost estimating relationship between the two variables, the cost and the weight. Also compute the **standard error and correlation coefficient** and give their **significance**. Plot the model estimated values and actual values of cost with respect to weight.

Q5.

Decide the best equipment to be selected for the cost and revenue data for four machines given below. Also do the ranking of the machines in terms of return being generated by them. Use **DELTA IRR/ERR** method (any of the two). **MARR = 18%, $\epsilon = 15\%$** .

(10)

EOY	Machine A	Machine B	Machine C	Machine D
0	-600000	-660000	-750000	-830000
1 2.31	210000	-30000	240000	260000
2 2.01	270000	330000	380000	360000
3 1.75	300000	350000	380000	440000
4 1.52	300000	350000	420000	440000
5 1.32	300000	400000	420000	440000
6 1.15	270000	400000	340000	350000
7	270000	340000	340000	350000

Q6.

A household is interested in installing 5KW rooftop solar system whose initial cost including the batteries, panels and installation is Rs 4,50,000. The system would generate 20 units (KWh) of electricity per day for entire year. The maintenance cost of the system is Rs 7500 per year and the batteries replacement costs Rs 1,00,000 every five years. The life of the system is 18 years and disposal value is Rs 50,000. Taking MARR of 12% per year, compute the net PW of the system. Assume cost of electricity is Rs 7 per unit. How much should be the initial investment to make the new PW Zero, if the new PW comes negative.

(10)

OR

Explain the differences among Design to Cost and Design to Price approaches of product costing in tabular form. Discuss the role of Value Engineering in achievement of target cost by drawing the flowchart of the target costing process. Also list the different activities performed under value engineering to improve the value of the product giving suitable examples for each of the activities.

(10)